

No. of Functional Features Included in the Solution

Date	04 July 2026
Team ID	SOLO-CCFD-2026
Project Name	Credit Card Fraud Detection using Machine Learning
Maximum Marks	5 Marks

Functional Features Overview

S.No	Feature Name	Description	Module	Status
1	Data Preprocessing	Scales Time/Amount, keeps PCA features, splits train/test	model	Done
2	SMOTE Balancing	Oversamples minority fraud class in training data only	model	Done
3	Multi-Model Training	Trains Logistic Regression, Decision Tree, Random Forest, XGBoost	model	Done
4	Model Comparison & Selection	Compares models by Precision/Recall/F1/ROC-AUC/AUPRC, picks best	model	Done
5	Single-Transaction Prediction	Web form returns fraud/genuine + probability instantly	flask_app	Done
6	Batch CSV Screening	Upload CSV of transactions, get predictions for all rows	flask_app	Done
7	JSON API Endpoint	/api/predict for programmatic integration	flask_app	Done
8	IBM Watson ML Deployment	Script to publish & deploy the model to a hosted endpoint	model	In Progress

Feature Summary

Metric	Count/Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	6 (preprocessing, SMOTE, training, comparison, single-prediction, batch)
Additional / Nice-to-Have Features	2 (JSON API, Watson ML deployment)
Features Tested & Verified	7